

for statistical modelling and machine learning, such as dimensionality reduction, clustering, regression, and classification, are available in scikit-learn. With extensive documentation and community support, it enables developers to efficiently use machine learning techniques in real-world problems, making it an ideal choice for both beginners and experienced practitioners.

Key features

- *Simple and consistent API:* scikit-learn offers a straightforward and consistent API across various algorithms, enabling easy experimentation and implementation.
- *Wide range of machine learning algorithms:* With a comprehensive selection of algorithms including regression, classification, clustering, and more, scikit-learn provides versatile solutions for diverse machine learning tasks.
- *Efficient data pre-processing and model evaluation:* scikit-learn facilitates seamless data pre-processing and model evaluation with its robust tools, enhancing the efficiency and effectiveness of machine learning workflows.

Limitations

- Limited support for deep learning
- Limited support for GPU acceleration

OpenCV: Unlocking the power of computer vision

OpenCV, short for Open Source Computer Vision, is a powerful open source library for computer vision and image processing tasks. Originally developed by Intel and now maintained by a global developer community, OpenCV is written in C++ with Python bindings, making it accessible to Python developers.

One of OpenCV's key strengths is its extensive collection of

algorithms and functions for image processing, computer vision, and machine learning. It offers capabilities such as image and video analysis, object detection and tracking, facial recognition, and feature extraction.

OpenCV's versatility is highlighted by its support for various platforms and operating systems, making it widely used in research and industry. Industries such as robotics, surveillance, augmented reality, medical imaging, automotive, and more leverage OpenCV for tasks like object detection, facial recognition, quality control, and healthcare image analysis.

Key features

- OpenCV offers a wide range of computer vision algorithms and functions for tasks like image processing, object detection, tracking, facial recognition, and feature extraction, making it suitable for various research and industry applications.
- *Cross-platform and language support:* OpenCV is designed to be cross-platform and supports multiple programming languages, including Python, C++, and Java.

Limitations

- Complexity and steep learning curve
- Limited documentation and community support

PyTorch: Dynamic deep learning for flexible innovation

PyTorch, developed primarily by Facebook's AI Research (FAIR) lab, is a widely used open source deep learning library known for its flexibility, efficiency, and ease of use. Its dynamic computation graph mechanism allows for intuitive model building and experimentation. This library is particularly beneficial for dynamic model architectures like recurrent neural networks (RNNs)

and transformers.

PyTorch's dynamic computational graph feature enables developers to define and modify computational graphs on-the-fly, unlike static computation graphs in other frameworks. This flexibility, combined with its imperative programming paradigm and seamless integration with Python, appeals to researchers and practitioners.

PyTorch is applied across various domains, including computer vision, natural language processing, and scientific computing. TensorFlow, developed by Google, is a primary alternative to PyTorch, offering a more static computation graph model that may be advantageous for certain applications, especially in production settings where performance and scalability are critical.

Keras, a high-level neural networks API, can run on top of TensorFlow, Microsoft Cognitive Toolkit, Theano, or PlaidML. It provides a simpler, more user-friendly interface compared to PyTorch, suitable for rapid prototyping and experimentation.

Key features

- Dynamic computation graph enables flexible and intuitive model design.
- *Pythonic interface:* Familiar and easy-to-use for Python developers.
- *Seamless integration with hardware accelerators:* GPU support for efficient training of deep neural networks.
- *Extensive ecosystem:* Rich set of libraries for computer vision, NLP, reinforcement learning, and more.

Limitations

- *Performance:* While PyTorch offers high flexibility and ease of use, it may not be as optimised for performance as some other frameworks, especially for large scale distributed training.
- *Production deployment:* Setting up